

Q1 2021 (01 Sep Shift 2)

A glass tumbler having inner depth of 17.5 cm is kept on a table. A student starts pouring water ($\mu = 4/3$) into it while looking at the surface of water from the above. When he feels that the tumbler is half filled, he stops pouring water. Up to what height, the tumbler is actually filled ?

- (1) 11.7 cm
- (2) 10 cm
- (3) 7.5 cm
- (4) 8.75 cm

Q2 2021 (27 Aug Shift 1)

In Millikan's oil drop experiment, what is viscous force acting on an uncharged drop of radius 2.0×10^{-5} m and density $1.2 \times 10^3 \text{ kg m}^{-3}$? Take viscosity of liquid $= 1.8 \times 10^{-5} \text{ N s m}^{-2}$. (Neglect buoyancy due to air).

- (1) 3.8×10^{-11} N
- (2) 3.9×10^{-10} N
- (3) 1.8×10^{-10} N
- (4) 5.8×10^{-10} N

Q3 2021 (26 Aug Shift 1)

A soap bubble of radius 3 cm is formed inside the another soap bubble of radius 6 cm. The radius of an equivalent soap bubble which has the same excess pressure as inside the smaller bubble with respect to the atmospheric pressure is cm.

Q4 2021 (26 Aug Shift 1)

Two narrow bores of diameter 5.0 mm and 8.0 mm are joined together to form a U-shaped tube open at both ends. If this U-tube contains water, what is the difference in the level of two limbs of the tube.

[Take surface tension of water $T = 7.3 \times 10^{-2} \text{ N m}^{-1}$, angle of contact $= 0$, $g = 10 \text{ ms}^{-2}$ and density of water $= 1.0 \times 10^3 \text{ kg m}^{-3}$]

(1) 3.62 mm

(2) 2.19 mm

(3) 5.34 mm

(4) 4.97 mm

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Answer Key

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Q1 (2) Q2 (2) Q3 (2) Q4 (2)
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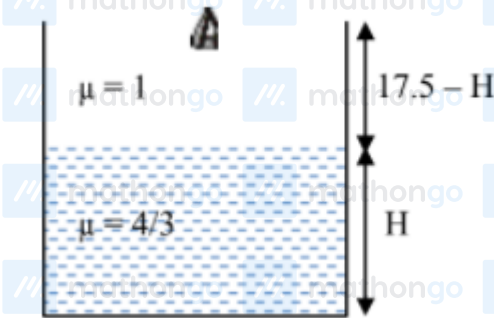
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Q1 (2)



Height of water observed by observer

$$= \frac{H}{\mu_w} = \frac{H}{(4/3)} = \frac{3H}{4}$$

Height of air observed by observer = $17.5 - H$

According to question, both height observed by observer is same.

$$\frac{3H}{4} = 17.5 - H \Rightarrow H = 10 \text{ cm Option (2)}$$

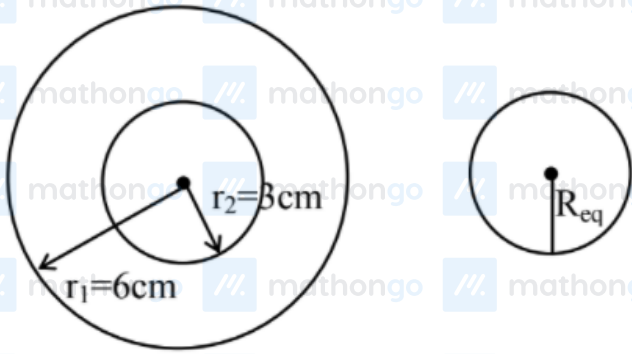
Q2 (2)

Viscous force = Weight

$$= \rho \times \left(\frac{4}{3} \pi r^3 \right) g$$

$$= 3.9 \times 10^{-10}$$

Q3 (2)



Excess pressure inside the smaller soap bubble

$$\Delta P = \frac{4S}{r_1} + \frac{4S}{r_2}$$

The excess pressure inside the equivalent soap bubble

$$\Delta P = \frac{4S}{R_{eq}}$$

From (i) & (ii)

$$\frac{4S}{R_{eq}} = \frac{4S}{r_1} + \frac{4S}{r_2}$$

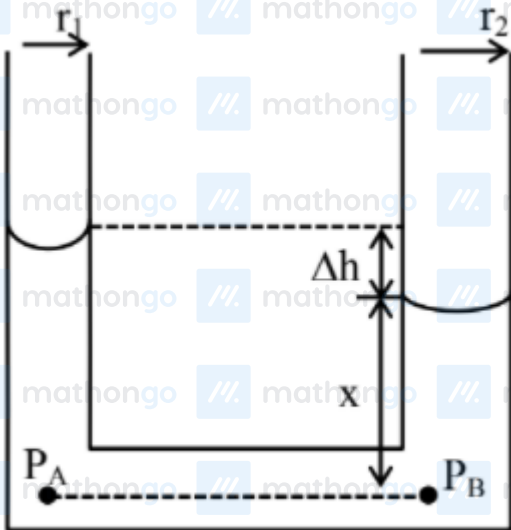
$$\frac{1}{R_{eq}} = \frac{1}{r_1} + \frac{1}{r_2}$$

$$= \frac{1}{6} + \frac{1}{3}$$

$$R_{eq} = 2 \text{ cm}$$

Ans. 2.00

Q4 (2)



We have $P_A = P_B$. [Points A & B at same horizontal level]

$$\therefore P_{\text{atm}} - \frac{2T}{r_1} + \rho g(x + \Delta h) = P_{\text{atm}} - \frac{2T}{r_2} + \rho g x$$

$$\therefore \rho g \Delta h = 2T \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$

$$= 2 \times 7.3 \times 10^{-2} \left[\frac{1}{2.5 \times 10^{-3}} - \frac{1}{4 \times 10^{-3}} \right]$$

$$\therefore \Delta h = \frac{2 \times 7.3 \times 10^{-2} \times 10^3}{10^3 \times 10} \left[\frac{1}{2.5} - \frac{1}{4} \right]$$

$$= 2.19 \times 10^{-3} \text{ m} = 2.19 \text{ mm}$$

Hence option (2)